ZAID KAMIL

Software Engineer, Unity Engine & XR

Los Angeles, CA | 424-382-6112 | zaidkamil@zrmap.com | linkedin.com/in/zaid-kamil/ | mzaidk.com

EDUCATION

California State University - Dominguez Hills | **M.S. in Computer Science** | Los Angeles, CA | Dec 2025

- **GPA: 3.90/4.00** | Focus on Real-Time Graphics, Distributed Systems, and XR Software Architecture

Texas A&M University | **B.S. in Chemical Engineering** | College Station, TX | Jul 2020

WORK EXPERIENCE

z/R Map | Software Engineer | Los Angeles, CA | Sep 2024 – May 2026

- **Engineered** custom **C# Unity engine extensions** for digital twin stadium visualizations, integrating real-time traffic and dynamic 3D environments.
- **Designed** GPU-accelerated rendering pipelines in Python, streaming live geospatial data to lower end-to-end latency by **40% at 90% uptime**.
- **Optimized** rendering utilizing **Scriptable Render Pipeline (SRP)** and custom **HLSL/GLSL shaders** to sustain **90 FPS** in real-time XR.
- **Architected** spatial synchronization services using **Netcode for GameObjects** to maintain stable state replication across 4 VR users.

Toro Auxiliary Partners | AR/VR Software Engineer | Los Angeles, CA | Jul 2024 – Present

- **Developed** interactive XR learning applications featuring custom **C# Unity plugins** for pulley system dynamics, box-push force vector visualizations, and spatial physics simulations.
- **Implemented** complex **3D vector math** and spatial transformations across a 3,000-line codebase to compute real-time raycasting, force vectors, and interactive mechanics.
- **Built** a multiplayer spatial networking service in Unity/C# using **Netcode**, enabling frame-accurate state replication and low-latency interaction for up to 10 concurrent users.
- **Enhanced** rendering quality and asset performance within Unity using **RenderDoc** and **Unity Profiler**, resulting in a **70% reduction** in frame drops across interactive physics lectures.

MindHome Inc. | Software Engineer Intern | Denver, CO | May 2024 – Nov 2024

- **Constructed** a safety-critical VR teleoperation app in **native C++ (P/Invoke) and C# (Unity, ROS)** featuring real-time hand tracking and haptic input compliant with **OpenXR**.
- **Designed** automated testing rigs and **CI/CD validation tools**, elevating automated test coverage to **85%** across 5 distinct VR device configurations.
- **Programmed** cross-hardware compatibility interfaces to resolve runtime input differences between HMD peripherals, reducing QA debugging time by **50%** per release.
- **Optimized** real-time spatial rendering and hand-tracking input pipelines targeting **Vulkan** and **OpenGL ES**, lowering overall application input-to-display latency to under **30ms**.

Division of IT, CSUDH | Information Technology Assistant | Los Angeles, CA | Feb 2022 – May 2026

- **Maintained** campus-wide network infrastructure across 150+ endpoints supporting hybrid learning, sustaining **90%+ system uptime** and resolving defects with a **95% satisfaction rate**.

Qatar Petroleum | Chemical Engineer | Doha, QA | Sep 2020 – Sep 2021

- **Automated** data reporting workflows in a safety-critical plant, saving **20 hours** weekly.

PROJECT EXPERIENCE & PUBLICATIONS

IoT Data Integration to AR/VR HMD | Lead Developer | Los Angeles, CA | Sep 2024 – Dec 2025

- **Designed** a distributed IoT telemetry pipeline (Azure IoT Hub, Stream Analytics, REST APIs) streaming live sensor data into XR environments for sub-second physical-world state visualization on HMD displays.

Publications:

- **Kamil, M.Z.** et al. (2020). *"Educational Mixed Reality Game on Water Desalination Plants."* IEEE.
- **Kamil, Z.** (2025). *"Real-Time Data Integration for Physical Environment Visualization in XR."* CSU Scholar.

SKILLS & INTERESTS

- **Languages & Low-Level APIs**: C++, C#, Python, Java, JavaScript, SQL, HLSL/GLSL, Vulkan, OpenGL ES, P/Invoke
- **Game Engines & XR SDKs**: Unity Engine (URP/HDRP, SRP, Netcode, Shader Graph), Unreal Engine, OpenXR, Meta Quest SDK, Mixed Reality Toolkit (MRTK), Lens Studio, ROS, WorldViz Vizard
- **3D Graphics & Performance**: RenderDoc, Unity Profiler, Memory Profiling, 3D Vector Math, Custom Render Passes, Native C++ Plugins, C# Engine Extensions, Real-Time Graphics, Low-Latency Optimization
- **Cloud & DevOps Tools**: Azure IoT Hub, AWS, Google Cloud, Docker, REST APIs, CI/CD, Git, Cisco LoRaWAN